# A1.1 Water (HL)

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 20 |

**About 20 minutes.** The number of marks shows how many separate points to make.

## Section A: Recall

5 MARKS

1 DEFINE the Goldilocks (habitable) zone. [A1.1.8] [1]

2 STATE two reasons why Earth has kept its water. [A1.1.7] [2]

3 NAME the type of meteorite thought to have delivered water to Earth. [A1.1.7] [1]

4 STATE how a deuterium atom differs from an ordinary hydrogen atom. [A1.1.7] [1]

## Section B: Understanding

7 MARKS

5 EXPLAIN why most of Earth's water is thought to have come from outside the planet. [A1.1.7] [2]

6 EXPLAIN how the ratio of deuterium to hydrogen is used to test where Earth's water came from. [A1.1.7] [2]

7 EXPLAIN why the search for life on other planets begins with a search for water. [A1.1.8] [3]

## Section C: Working with evidence

8 MARKS

*Approximate values for four bodies in our solar system.*

**8** [A1.1.7, A1.1.8]

| BODY | DISTANCE FROM SUN / AU | MEAN SURFACE TEMPERATURE / °C | RADIUS / EARTH = 1 | WATER |
|---|---|---|---|---|
| VENUS | 0.7 | 464 | 0.95 | Vapour only |
| EARTH | 1.0 | 15 | 1.00 | Liquid oceans |
| MARS | 1.5 | −63 | 0.53 | Ice; traces of ancient rivers |
| EUROPA (MOON OF JUPITER) | 5.2 | −160 | 0.25 | Ice crust; probable ocean beneath |

The Sun's habitable zone is roughly 0.95 to 1.7 AU.

a **IDENTIFY** the bodies that lie inside the habitable zone. [1]

b **EXPLAIN** why Venus has no liquid water. [2]

c **SUGGEST** why Mars has no liquid water on its surface today, even though it is in the habitable zone. [2]

d **SUGGEST** how Europa could have liquid water although it is far outside the habitable zone. [2]

e **OUTLINE** what Europa shows about the limits of the habitable zone idea. [1]

What the Cell. Original questions written for this course.